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Fast Commutation Error Compensation Method of Sensorless Control for MSCMG BLDC Motor with Non-ideal Back-EMF

Haifeng Zhang and Haitao Li

***Abstract*—As aerospace applications, high reliability and low power consumption operation of the brushless direct current (BLDC) motor for the magnetically suspended control moment gyro (MSCMG) is the primary consideration. To guarantee reliability, the sensorless control without position sensors is adopted. Due to the phase lag caused by hardware and software, the commutation error is inevitable. In order to reduce losses, this paper proposes a fast commutation error compensation method to improve commutation accuracy. First, the control topology of the BLDC motor with a Buck converter and the commutation points detection circuit are presented. Then, the conducting back electromotive force (EMF) contained in a novel derived DC-link voltage expression is estimated through an extended state observer (ESO). Finally, the commutation error is extracted from the conducting back-EMF for compensation. Different from the conventional closed-loop correction method based on the back-EMF symmetry, the proposed commutation error compensation method can obtain the commutation error within only one commutation cycle by a continuous estimation with the ESO. The effectiveness of the proposed fast commutation error compensation method was evaluated with an MSCMG system.**

***Index Terms*—Sensorless control, brushless direct-current (BLDC) motor, extended state observer (ESO), commutation error compensation, magnetically suspended control moment gyro (MSCMG)**

I. Introduction

CONTROL moment gyro (CMG) is an attitude control device generally used in spacecraft attitude control systems [1]. As the rotor tilts, the changing angular momentum causes a gyroscopic torque that rotates the spacecraft [2, 3]. Because the magnetically suspended bearing has the characteristics of zero friction, no lubrication required and very slight vibration, the CMG equipped with a magnetically suspended bearing system (MSCMG) can achieve longer service life and higher control precision [4, 5].

Since the brushless direct-current (BLDC) motor has the advantages of high-power density and high efficiency, it is an ideal driver for the high-speed rotor system of the MSCMG [6]. Three switch Hall effect sensors are usually mounted on the stator to provide six discrete commutation signals for the BLDC motor. However, it occupies the installation space and reduces the reliability of the motor [7, 8], and is also affected by the installation accuracy and operating environment [9]. Therefore, the sensorless control strategy for the BLDC motor provides an ideal solution to overcome these problems. Various sensorless control methods have been proposed and successfully applied to different motor systems [10–12]. Researchers generally focus on two issues of the sensorless control, one is to achieve reliable driving without position sensors, and the other is to improve commutation accuracy.

For the first issue, numerous methods have been proposed to detect the commutation points, where the back electromotive force (EMF) based methods are commonly used [6]. In [13], a sensorless control scheme was described by detecting the zero-crossing point (ZCP) of the terminal voltage from unconducting windings to estimate the rotor position. For the motor that is difficult to wire out the neutral line, Cui et al. utilized a virtual resistance network to construct the neutral point voltage [14], and P. Damodharan et al. obtained the commutation points by detecting the line-line voltage difference [15], which eliminates the requirement for the neutral line. To compensate for the phase lag caused by the low-pass filter, Chun et al. proposed a novel phase compensation method based on a hysteresis comparator, which can also prevent multiple output transitions from noise or ripple in terminal voltages [16]. Different from the methods based on the ZCPs of the back-EMF, some new commutation point detection methods have also been proposed. In [11], a new sensorless commutation method for a high-speed BLDC motor was proposed, and it detected the commutation point by a speed-independent flux-linkage function. Based on the constant relationship between the third harmonic of the back-EMF and the rotor flux-linkage, Song et al. developed an active compensation method based on the virtual third harmonic back-EMF to detect the commutation point accurately [17].

Due to the efforts of researchers on the sensorless control, reliable driving of the BLDC motor without position sensors is guaranteed. However, because of inaccurate hardware parameters, variable operating conditions and other un-modeled factors, the commutation error is inevitable [18], which results in low efficiency and high losses of the motor [19]. Therefore, for the second issue, the commutation error problem has been receiving more and more attention, and many studies have been devoted to improving commutation accuracy. In [20], the

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commutation error caused by replacing the back-EMF with the terminal voltage was considered, and an error compensation circuit to ensure proper commutation was proposed. Based on the symmetric terminal voltages of the non-conducting phase at the beginning and end of the commutation cycle, Jang et al. presented a method to detect the improved commutation point [21]. In [18, 22], the integration difference between the adjacent 60° interval before and after the commutation was used to compensate for the commutation error. In [9], the relationship between the commutation error and the phase current deviation was derived, and a closed-loop commutation error compensation method based on the relationship was proposed. Li et al. analyzed the relationship between commutation error and DC-link current, and proposed a model-free adaptive control method to compensate for the commutation error [23]. In [24], Gu et al. considered the total commutation error as the internal power factor angle and achieved the commutation error correction by regulating the angle to zero. As presented, various methods have been proposed and applied to different BLDC motor systems to solve the commutation error problems. Therefore, the characteristics of different motor systems should be fully considered to develop the commutation error compensation method that is more appropriate for different systems.

Different from the typical BLDC motor stator structure, as an aerospace application, the MSCMG BLDC motor used in this paper adopted an iron-less stator structure to reduce the core losses [25]. Without iron core in the stator, the inductance of the stator winding presented to be very small. If the three-phase bridge is modulated through the pulse width modulation (PWM), it would cause a severe modulation noise, leading to a large number of high-order harmonics in the current [23]. The high-order harmonics would aggravate skin effect to increase high-frequency copper losses [26] and also increase eddy current losses in the rotor [27]. In order to reduce those losses, a control circuit structure with a Buck converter added in front of the three-phase inverter was proposed to deal with the modulation noise, especially for small inductance motors [28]. Because the Buck converter can adjust the DC-link voltage, the three-phase bridge only performs commutation operation without modulation [18]. Therefore, rare modulation noise would be introduced to the terminal voltage and the phase current, which helps reduce the losses of the motor [19].

Another difference of the MSCMG BLDC motor is its air-gap. The air-gap of the magnetically suspended BLDC motor is much wider than that of the motor suspended by the mechanical bearing [6, 19]. Fig. 1 shows the air-gap magnetic field distribution of the MSCMG BLDC motor with an iron-less stator. As can be seen, when the rotor is suspended in the middle of the air-gap, the waveform of the air-gap flux density is not an ideal trapezoidal waveform and more approaches a sinusoidal waveform [29].

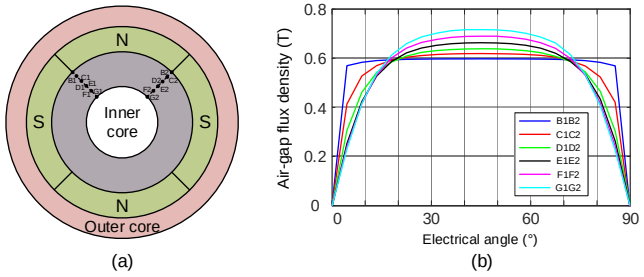


Fig. 1. Air-gap magnetic field distribution of the MSCMG BLDC motor with iron-less stator. (a) Marks of the air-gap position. (b) Waveforms of the air-gap flux density.

As presented, most compensation or correction methods can not obtain the commutation error, but regulate terminal voltage [6, 12, 17, 18, 20, 29, 30], phase current [9, 22] or DC-link current [19, 23, 31] to be symmetrical. Therefore, several commutation cycles are needed to complete the compensation process. The main contribution of this paper is that it can estimate the commutation error within only one commutation cycle through an extended state observer (ESO) based on a novel derived DC-link voltage expression. The rest of this paper is organized as follows. Section II gives the sensorless control method and analyzes the commutation process. Section III derives the expression of the conducting back-EMF containing the commutation error and describes the design issues of the ESO. Then, with the conducting back-EMF, the commutation error detection and compensation processes are presented. Detailed experimental tests of the proposed fast commutation error compensation method are performed on an MSCMG BLDC motor system in Section IV. Some important conclusions are summarized in Section V.

II. Sensorless Control and Commutation Process Analysis of MSCMG BLDC Motor

A. Sensorless Control of MSCMG BLDC Motor

Fig. 2 shows the control topology of the MSCMG BLDC motor. As can be seen, different from the conventional control topology of the BLDC motor, a Buck converter is added in front of the three-phase bridge. The Buck converter consists of a switching transistor T_B, together with a inductor L_b, a capacitor C_b and a diode D_b. When T_B is switched on, the power supply u_{in} is supplying current to the motor, and the energy is also stored in L_b and C_b. When T_B is switched off, the energy stored in the magnetic filed around L_b is released back into the circuit. Once L_b has returned a large part of its stored energy to the circuit, the charge stored in C_b becomes the main source of current until the next cycle. The switching transistor T_B is modulated by the PWM, where the switching frequency is 20 kHz. Because the Buck converter can step down the voltage from power supply, the acceleration and deceleration operations of the BLDC motor can be controlled

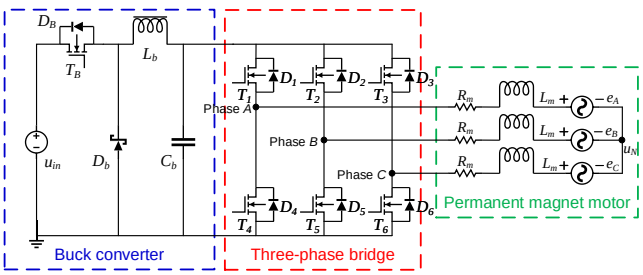


Fig. 2. Control topology of the MSCMG BLDC motor.

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by adjusting the duty cycle of the pulse width. With the Buck converter, no PWM is needed to perform on the three-phase bridge. The Bridge only performs commutation operation, and rare modulation noise would be introduced to the phase current and the terminal voltage. The inductance L_b and the capacitance C_b in the Buck converter are 0.3 mH and 5 uF, respectively. The DC-link voltage output by the Buck converter is $u_m = Du_{in}$, where D is the duty cycle controlled by T_B.

The MSCMG BLDC motor adopted a surface-mounted permanent magnet structure, and there is no change in the rotor inductance with angle. The terminal voltage equations of the BLDC motor are presented as

$$u_x = R_m i_x + L_m \frac{di_x}{dt} + e_x + u_N \tag{1}$$

where $x = A, B, C$; u_x, i_x and e_x are the terminal voltage, phase current and back-EMF of phase x, respectively; R_m is the stator phase resistance; L_m is the stator phase inductance; u_N is the neutral point voltage. The back-EMFs of the BLDC motor are given as a form of the Fourier series

$$e_x = K_e \omega_e \sum_{m=1}^{\infty} A_{xm} \cos\left(m\left(\theta_e + n\frac{2\pi}{3}\right)\right) \tag{2}$$

where $n = 0, -1, 1$ when $x = A, B, C$, respectively; K_e is the back-EMF coefficient; ω_e is the electrical angular velocity; θ_e is the electrical rotor position; A_{xm} is the amplitude of mth harmonic in the phase x.

According to the terminal voltages, Fig. 3 shows the commutation points detection circuit of sensorless control for the MSCMG BLDC motor. Limited by the number of contact terminals of the aviation connector plug of the MSCMG BLDC motor, there is no reserved neutral line. Therefore, ZCPs of the line-line voltage are used as the commutation signals that are also called virtual Hall signals, H_A, H_B, H_C. As analyzed, the terminal voltage contains no modulation noise because of the Buck converter, and the cut-off frequency of the low-pass filter in Fig. 3 can be appropriately increased.

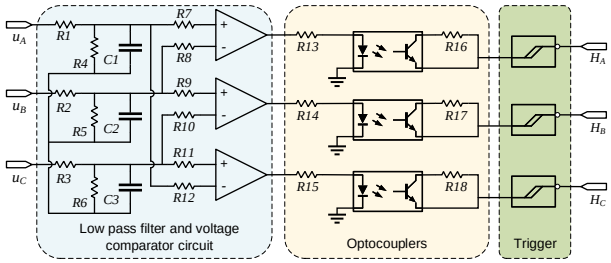


Fig. 3. Commutation points detection circuit of sensorless control for the MSCMG BLDC motor.

B. Commutation Process Analysis

Fig. 4 shows the relationship between virtual Hall signals and back-EMFs. The following takes the commutation from region III to region IV for example to analyze the commutation process. As shown in Fig. 5, before commutation, the current flows from Path2 to Path4. After commutation, the transistor T_2 turns off and T_3 turns on. For the existence of the inductance in the stator winding, the current in phase B can not decrease to zero immediately, as shown in Fig. 6, it will

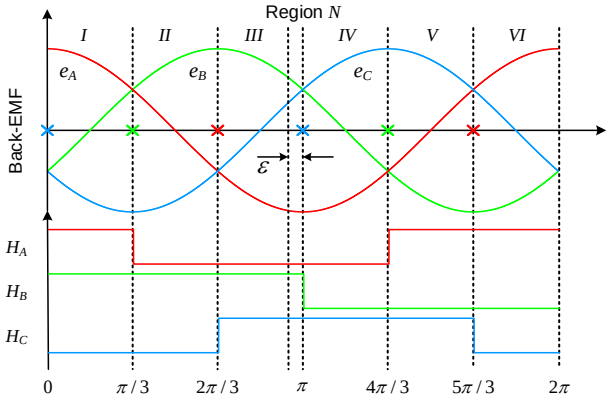


Fig. 4. Relationship between virtual Hall signals and back-EMFs.

experience a short freewheeling process, and the current flows from Path1 and Path5 to Path 4, as shown in Fig. 5. Therefore, during the freewheeling process, the phase B is shorted to the ground, and $u_B = 0$. Since $u_A = 0$, $u_C = u_m$, according to (1), the neutral point voltage is

$$u_N = \frac{u_m - e_A - e_B - e_C}{3} \tag{3}$$

and the phase currents are obtained through solving the differential equation of (1)

$$\begin{cases} i_A(t) = -\dfrac{v_A}{R_m} + \left(\dfrac{v_A}{R_m} - I_m\right) e^{-\frac{R_m}{L_m}t} \\ i_B(t) = -\dfrac{v_B}{R_m} + \left(\dfrac{v_B}{R_m} + I_m\right) e^{-\frac{R_m}{L_m}t} \\ i_C(t) = -\dfrac{v_C}{R_m} + \dfrac{v_C}{R_m} e^{-\frac{R_m}{L_m}t} \end{cases} \tag{4}$$

where $v_A = (u_m + 2e_A - e_B - e_C)/3$, $v_B = (u_m - e_A + 2e_B - e_C)/3$ and $v_C = (-2u_m - e_A - e_B + 2e_C)/3$. It can be seen from (4) that i_B will decrease to zero, and i_C will increase to the normal current I_m, as shown in Fig. 6. As analyze in [32], because the commutation interval mainly decided by the inductance, it is feasible to neglect the effort for small inductance motors.

1) Analysis of Ideal Commutation Process: The ideal commutation points are the intersections of the phase back-EMFs. After the freewheeling process is finished, the current flows from phase C to phase A. Neglecting the commutation interval, $e_B = e_C$ before and after commutation, as shown in

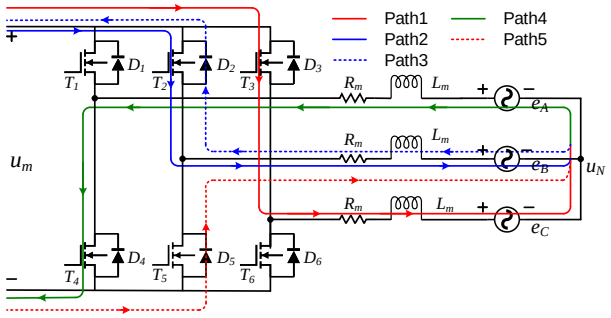


Fig. 5. Commutation process when phase B transfers to phase C.

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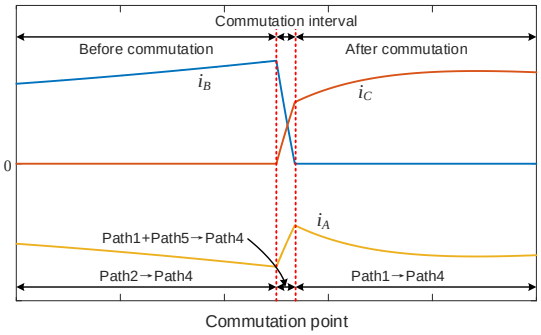


Fig. 6. Diagram of phase current before and after commutation.

Fig. 4. The electromagnetic torques before and after commutation are

$$\begin{cases} T_{e1} = \dfrac{e_A i_A + e_B i_B}{\omega_r} \\ T_{e2} = \dfrac{e_A i_A + e_C i_C}{\omega_r} \end{cases} \tag{5}$$

where ω_r is mechanical angular velocity. Since $i_B = -i_A$ before commutation and $i_C = -i_A$ after commutation, it is clear that $T_{e2} = T_{e1}$. It implies that it can achieve a continuous torque before and after commutation.

2) Analysis of Advanced/Lagged Commutation Process: As analyzed in [30], due to the low-pass filter lag, device delay and software delay, it inevitably leads to a commutation error. If the commutation error is large enough, the abnormal freewheeling will occur, which increases losses of the motor.

As shown in Fig. 5, after commutation, the current flows from Path1 to Path4, and $i_C = -i_A$. Define the voltage drop caused by the phase resistance and inductance as

$$u_{RL} = R_m i_C + L_m \frac{di_C}{dt} = -R_m i_A - L_m \frac{di_A}{dt} \tag{6}$$

and the DC-link and neutral point voltages are

$$\begin{cases} u_m = 2u_{RL} + e_C - e_A \\ u_N = u_m - u_{RL} - e_C \end{cases} \tag{7}$$

Therefore, the voltage between phase B and DC-link is

$$u_{Bm} = e_B - e_C - u_{RL} \tag{8}$$

Since the high-order harmonics of the back-EMF are very small, only the fundamental is taken into consideration. Substituting for e_B and e_C from (2) into (8), u_{Bm} is obtained as

$$\begin{aligned} u_{Bm} &= K_e\omega_e(\cos(\theta_e - 2\pi/3) - \cos(\theta_e + 2\pi/3)) - u_{RL} \\ &= \sqrt{3}K_e\omega_e \sin(\theta_e) - u_{RL} \end{aligned} \tag{9}$$

If the actual commutation point advances the ideal commutation point by ε, the commutation point is $\theta_e = \pi - \varepsilon$, as shown in Fig. 4. Substituting $\theta_e = \pi - \varepsilon$ in (9), it yields

$$u_{Bm} = \sqrt{3}K_e\omega_e \sin(\varepsilon) - u_{RL} \tag{10}$$

As can be seen, once u_{Bm} is larger than the instantaneous forward voltage of the diode D_2, there will exist an abnormal current flowing through the Path3, as shown in Fig. 5. It clear from (10) that u_{Bm} is not only related to the commutation error ε, but also is affected by speed and motor parameters. TABLE I gives the parameters of the MSCMG BLDC motor. To illustrate the condition of the abnormal freewheeling, Fig. 7 shows the relationship of u_{Bm} and the commutation error ε at different currents and speeds. Since the stator inductance is very small, its voltage drop is neglected in the analysis. As can be seen, u_{Bm} is positively related to commutation error ε and the motor speed, and a larger current may relieve the occurrence of the abnormal freewheeling. However, since the rotor of the MSCMG BLDC motor is suspended by active magnetically bearings (AMBs), it can operate at high speed with a small current under steady-state. Therefore, the abnormal freewheeling is especially noticeable in the MSCMG BLDC motor.

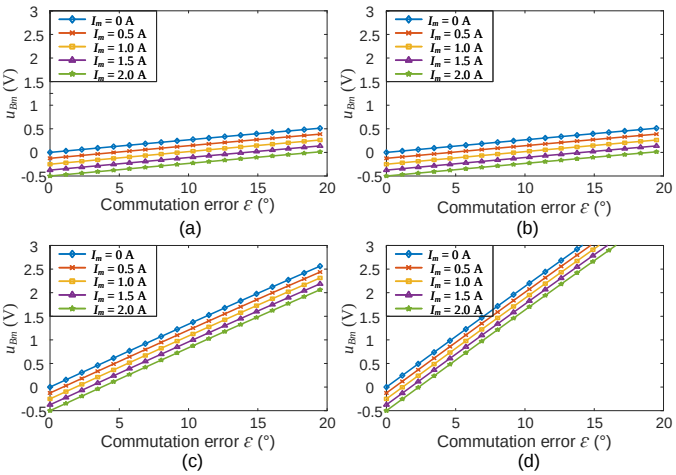


Fig. 7. Relationship of u_{Bm} and the commutation error at different currents and speeds. (a) 1000 r/min. (b) 3000 r/min. (c) 5000 r/min. (d) 8000 r/min.

Fig. 8 demonstrates the simulation results of phase currents and back-EMFs at 8000 r/min with ideal commutation, advanced commutation and lagged commutation. As shown in Fig. 8, if the commutation error is large enough, the abnormal freewheeling occurs, and the currents will be distorted. Since the back-EMF and the abnormal freewheeling current are in opposite directions, according to the electromagnetic torque equation

$$T_e = \frac{e_A i_A + e_B i_B + e_C i_C}{\omega_r} \tag{11}$$

a negative torque is produced by the abnormal freewheeling current, which reduces the efficiency of the motor and may cause cooling problems.

TABLE I
MOTOR PARAMETERS

Parameter	Unit	Value
Dc-link voltage	V	28
Loading rotating inertia	kg·m^2	0.13021
EMF coefficient	V/(r/min)	0.00088
Moment coefficient	N·m/A	0.0084
Phase resistance	Ω	0.25
Phase inductance	μH	7.5
Rated speed	r/min	9000
Rated current	A	5
Maximum current	A	8
Pole pairs		2

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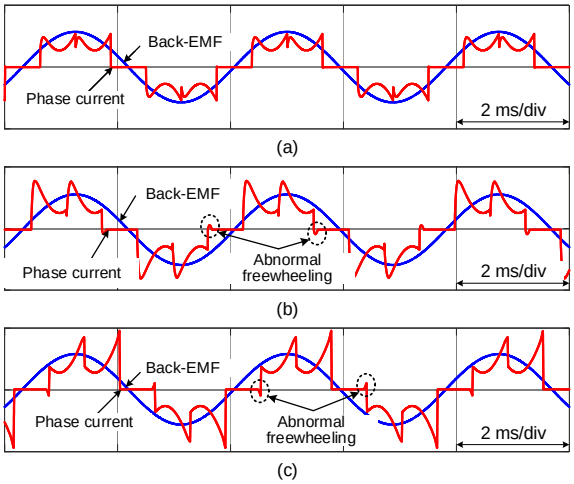


Fig. 8. Simulation results of the phase currents and the back-EMFs at 8000 r/min. (a) Ideal commutation. (b) Advanced commutation. (c) Lagged commutation.

It is noted that the previous analysis is based on the advanced commutation, and the characteristic of the lagged commutation can also be obtained according to the previous analysis. Therefore, if the commutation error is not corrected and compensated in time, it will lead to low efficiency, deteriorating the operation performance of the motor.

III. Principle and Implementation of Fast Commutation Error Compensation Method

A. Principle of Commutation Error Detection

Ideally, the waveform of the back-EMF of BLDC motors is trapezoidal [33]. However, as presented in Fig. 1, the waveform of the air-gap flux density of the MSCMG BLDC motor approaches sinusoidal. Fig. 9 shows the phase back-EMF and its fast Fourier transform (FFT) of the MSCMG BLDC motor around 3000 r/min. As can be seen, apart from the fundamental, the phase back-EMF also contains an amount of high-order harmonics. However, the amplitudes of the high-order components of the back-EMF, such as $3^{\rm rd}$ harmonic, $5^{\rm th}$ harmonic and $7^{\rm th}$ harmonic, are all very small and below -40 dB. Therefore, it is acceptable to neglect the high-order

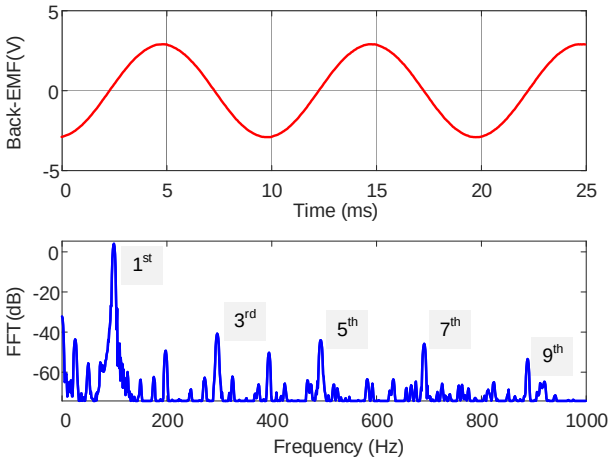


Fig. 9. Phase back-EMF and its FFT of the MSCMG BLDC motor.

components of the back-EMF in the following analysis, and from (2), it obtains

$$e_x = K_e\omega_e \cos\left(\theta_e + n\frac{2\pi}{3}\right) \tag{12}$$

As shown in Fig. 4, the ideal commutation points are 0, $\pi/3$, $2\pi/3$, π, $4\pi/3$ and $5\pi/3$. If there exists a commutation error, the actual commutation points will be ε_1, $\pi/3+\varepsilon_2$, $2\pi/3+\varepsilon_3$, $\pi+\varepsilon_4$, $4\pi/3+\varepsilon_5$ and $5\pi/3+\varepsilon_6$, where ε_{1-6} denotes the angle that the actual commutation point lags the ideal commutation point. In region I, $\theta_e \in [0, \pi/3)$, the current flows from phase A to phase C, and $i_A = i_m$, $i_C = -i_m$, where i_m is the DC-link current. The DC-link voltage equation is

$$u_m = 2R_m i_m + 2L_m\frac{di_m}{dt} + e_A - e_C \tag{13}$$

Substituting the back-EMFs model (12) into (13), it yields

$$u_m{=}2R_m i_m + 2L_m\frac{di_m}{dt} + \sqrt{3}K_e\omega_e\sin(\theta_e + \frac{\pi}{3}) \tag{14}$$

If there exists a commutation error ε_1, the DC-link voltage equation is rewritten as

$$u_m{=}2R_m i_m + 2L_m\frac{di_m}{dt} + \sqrt{3}K_e\omega_e\sin(\theta_e + \frac{\pi}{3} + \varepsilon_1) \tag{15}$$

Regarding $\theta_e + \pi/3$ as θ_{e1}, it obtained from (15)

$$u_m{=}2R_m i_m + 2L_m\frac{di_m}{dt} + \sqrt{3}K_e\omega_e\sin(\theta_{e1} + \varepsilon_1) \tag{16}$$

Similarly, the DC-link voltage equations in different regions are summarized as

$$u_m{=}2R_m i_m + 2L_m\frac{di_m}{dt} + \sqrt{3}K_e\omega_e\sin(\theta_{eN} + \varepsilon_N) \tag{17}$$

where $\theta_{eN} = \begin{cases} \theta_e + \pi/3, & N=1, \quad \theta_e \in [0, \pi/3) \\ \theta_e, & N=2, \quad \theta_e \in [\pi/3, 2\pi/3) \\ \theta_e - \pi/3, & N=3, \quad \theta_e \in [2\pi/3, \pi) \\ \theta_e - 2\pi/3, & N=4, \quad \theta_e \in [\pi, 4\pi/3) \\ \theta_e - \pi, & N=5, \quad \theta_e \in [4\pi/3, 5\pi/3) \\ \theta_e - 4\pi/3, & N=6, \quad \theta_e \in [5\pi/3, 2\pi) \end{cases}$.

From (17), it is concluded that θ_{eN} in different regions all belongs to $[\pi/3, 2\pi/3)$. Therefore, the DC-link voltage equation (17) can be rewritten as an unified formulation

$$u_m{=}2R_m i_m + 2L_m\frac{di_m}{dt} + f(\theta_{eu}, \varepsilon_u) \tag{18}$$

where $f(\theta_{eu}, \varepsilon_u) = \sqrt{3}K_e\omega_e\sin(\theta_{eu}+\varepsilon_u)$, $\theta_{eu} \in [\pi/3, 2\pi/3)$ is the conducting back-EMF, and ε_u is the commutation error. As analyzed previously, the commutation error is contained in $f(\theta_{eu}, \varepsilon_u)$. To illustrate the waveforms of $f(\theta_{eu}, \varepsilon_u)$ with different commutation condition, Fig. 10 shows the waveforms of $f(\theta_{eu}, \varepsilon_u)$ with ideal commutation, lagged commutation and advanced commutation. As can be seen, if the commutation error $\varepsilon_u > 0$, the commutation operation lags the ideal commutation point, and if the commutation error $\varepsilon_u < 0$, the commutation operation advances the ideal commutation point. As analyzed in [6], since the MSCMG BLDC motor has a very small torque coefficient and a flywheel rotor with a large moment of inertia, the speed changes slowly even under

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acceleration and deceleration, and approximates a constant during each commutation cycle. Therefore, it is feasible to extract the commutation error from $f(\theta_{eu}, \varepsilon_u)$.

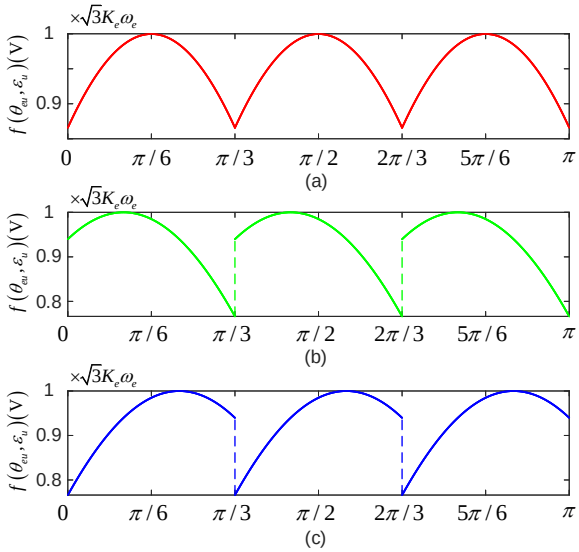


Fig. 10. Waveforms of $f(\theta_{eu}, \varepsilon_u)$ with ideal commutation, Lagged commutation and advanced commutation. (a) Ideal commutation. (b) Lagged commutation when ε_u = 10°. (c) Advanced commutation when ε_u = -10°.

B. Conducting Back-EMF Estimation with ESO

The ESO is the core component of active disturbance rejection control (ADRC). It expands the lumped disturbance into a new state variable of the system and then uses the input and output of the system to reconstruct all states that contain the original state variables and disturbances of the system. For the characteristic of its fast convergence, low overshoot, high observation precision and simple structure, it has been successfully applied in motor and other industrial control. The following describes the estimation process of the conducting back-EMF with the ESO.

According to (18), the DC-link voltage equation is rewritten as

$$\begin{cases} \dfrac{di_m}{dt} = \dfrac{1}{2L_m}u_m + d \\ d = -\dfrac{R_m}{L_m}i_m - \dfrac{1}{2L_m}f(\theta_{eu}, \varepsilon_u) \end{cases} \tag{19}$$

where d represents the lumped disturbance. It is clear from (19) that the commutation error ε_u is contained in d, and once d is estimated, ε_u can be obtained.

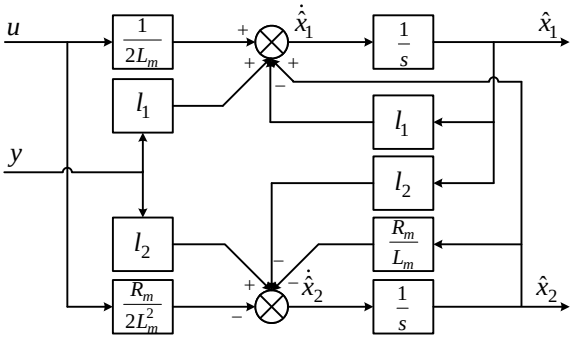


Fig. 11. Block diagram of the ESO.

Define state variables $x_1 = i_m$, $x_2 = d$, input variable $u = u_m$, and the extended state space form of the system (19) is given as

$$\begin{cases} \dot{x} = Ax + Bu + E\dot{f}(\theta_{eu}, \varepsilon_u) \\ y = Cx \end{cases} \tag{20}$$

where $A = \begin{bmatrix} 0 & 1 \\ 0 & -R_m/L_m \end{bmatrix}$, $B = \begin{bmatrix} 1/(2L_m) \\ -R_m/(2L_m^2) \end{bmatrix}$, $E = \begin{bmatrix} 0 \\ -1/(2L_m) \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$. Accordingly, the ESO is designed as

$$\begin{cases} \dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly \\ \hat{y} = C\hat{x} \end{cases} \tag{21}$$

where $L = [l_1 \quad l_2]^T$ is the error feedback gain matrix of the ESO, $\hat{x}$ and $\hat{y}$ are the estimated values of x and y, respectively. According to (20) and (21), the block diagram of the ESO is obtained, as shown in Fig. 11.

To configure the parameter, Gao proposed a practical parameter tuning method for the ESO, which simplifies the controller parameters tuning to a single parameter by making them a function of the loop-gain bandwidth [34]. Generally, the poles of the ESO are always placed at the same position. If the poles are placed at point $-\omega_0$, the characteristic equation of the ESO satisfies

$$\lambda(s) = |sI - (A - LC)| = (s + \omega_0)^2 \tag{22}$$

From (22), it yields

$$\begin{cases} l_1 = 2\omega_0 - \dfrac{R_m}{L_m} \\ l_2 = \omega_0^2 - \dfrac{2R_m}{L_m}\omega_0 + \dfrac{R_m^2}{L_m^2} \end{cases} \tag{23}$$

When the ESO converges, the lump disturbance d can be estimated, and $f(\theta_{eu}, \varepsilon_u)$ is

$$f(\theta_{eu}, \varepsilon_u) = -2(R_m i_m + L_m d) \tag{24}$$

C. Commutation Error Compensation

Since $f(\theta_{eu}, \varepsilon_u)$ has been obtained, the commutation error needs to be extracted from $f(\theta_{eu}, \varepsilon_u)$. Fig. 12 shows the waveform of $f(\theta_{eu}, \varepsilon_u)$. As can be seen, $f(\theta_{eu}, \varepsilon_u)$ is not monotonic, and one $f(\theta_{eu}, \varepsilon_u)$ maps two rotor positions, θ_{em}

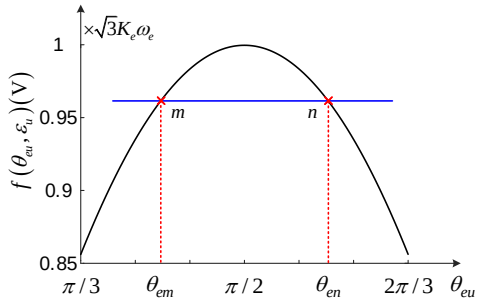


Fig. 12. Waveform of $f(\theta_{eu}, \varepsilon_u)$.

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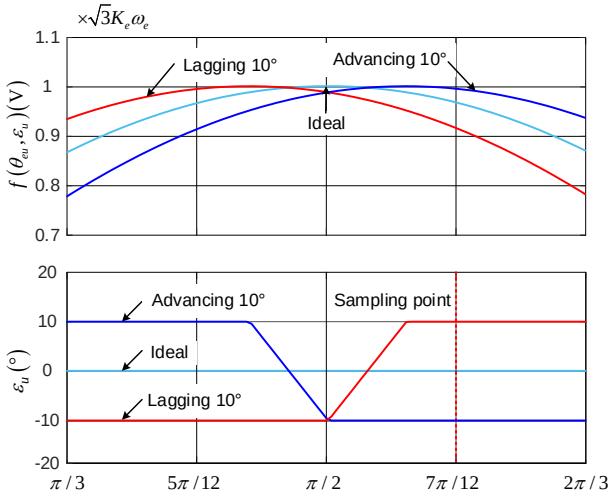


Fig. 13. Schematic of the commutation error extraction process.

and θ_{en}. To deal with the conflict in extracting the commutation error, a trigonometric function transformation is applied. Through the transformation, the estimated commutation error is

$$\hat{\varepsilon}_u = \arcsin\left(\sin\left(\theta_{eu}\right)\right) - \arcsin\left(\frac{f\left(\theta_{eu}, \varepsilon_u\right)}{\sqrt{3}K_e\omega_e}\right) \tag{25}$$

where θ_{eu} increases from $\pi/3$ to $2\pi/3$ during each commutation cycle. According to (25), the schematic of the commutation error extraction process is shown in Fig. 13. Since the commutation error is estimated by the ESO, it must undergo a convergence process even though it is very short. Therefore, the commutation error should be sampled after the ESO has converged, and it is better to select the sampling point in the second half of each commutation cycle. Meanwhile, the sampling point is also constrained by the max commutation error of the BLDC motor for it would miss the next ideal commutation point when the sampling point is too late. The main commutation error source is the phase lag caused by the low-pass filter to filter the phase back-EMF [9, 11, 14]. The cut-off frequency of the low-pass filter in this study is configured as 2 kHz, and the total phase lag is about 9° at

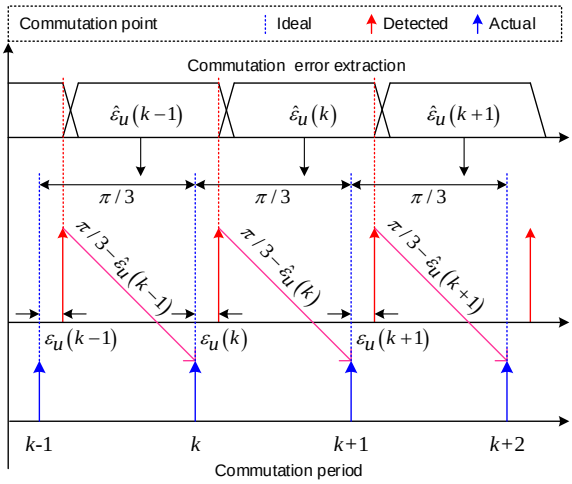


Fig. 14. Diagram of the commutation error compensation process.

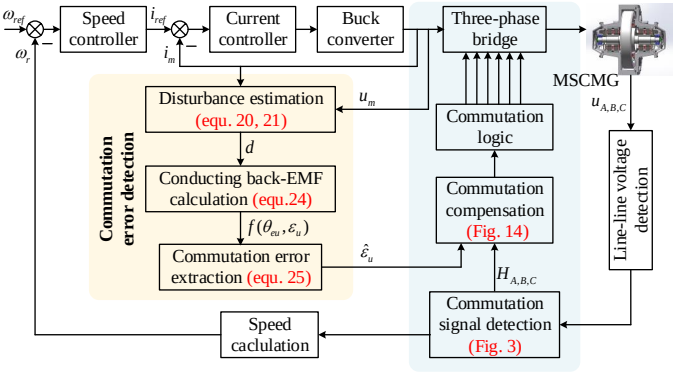


Fig. 15. Block diagram of the sensorless control with fast commutation error compensation method.

related speed. Therefore, the commutation error samples at $3/4$ of each commutation cycle.

The diagram of the commutation error compensation process is shown in Fig. 14. As can be seen, once the system detects the commutation point, it starts to perform the commutation error extraction algorithm, and after $\pi/4$, as shown in Fig. 13, the commutation error is sampled. As presented in Fig. 14, the dotted blue line is the ideal commutation point that is the ZCP of the line-line voltage theoretically. Due to the phase lag caused by the detection system, the detected commutation point (red arrowed line) has lagged the ideal commutation point. Therefore, the extracted commutation error can only be used for the next commutation point. For example, when the commutation point is detected around $k-1$, a phase delay of $\pi/3 - \hat{\varepsilon}_u$ should be made until performing commutation operation at k. It can be seen from the commutation error compensation process that the commutation error can be obtained within only one commutation cycle, and therefore the proposed method is much faster than other commutation compensation methods.

Based on the above-mentioned analysis, the block diagram of the sensorless control with the fast commutation error compensation method is shown in Fig. 15. As can be seen,

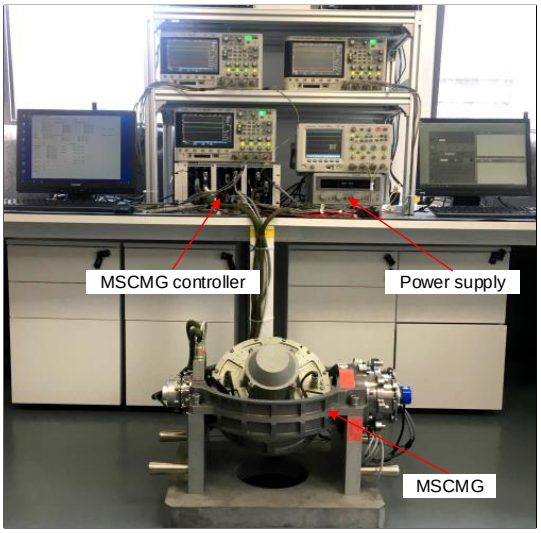


Fig. 16. Experiment platform of the MSCMG.

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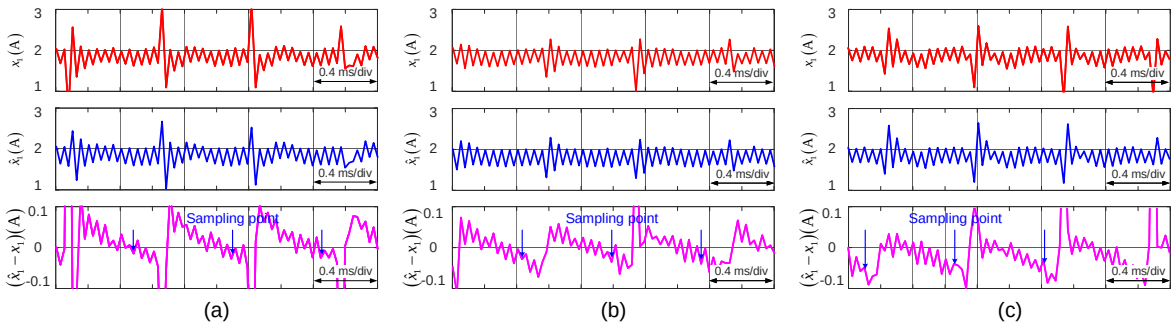


Fig. 17. Performance of the ESO to track the state variables x_1 at 9000 r/min. (a) Advanced commutation by 10°. (b) Normal commutation. (c) Lagged commutation by 10°.

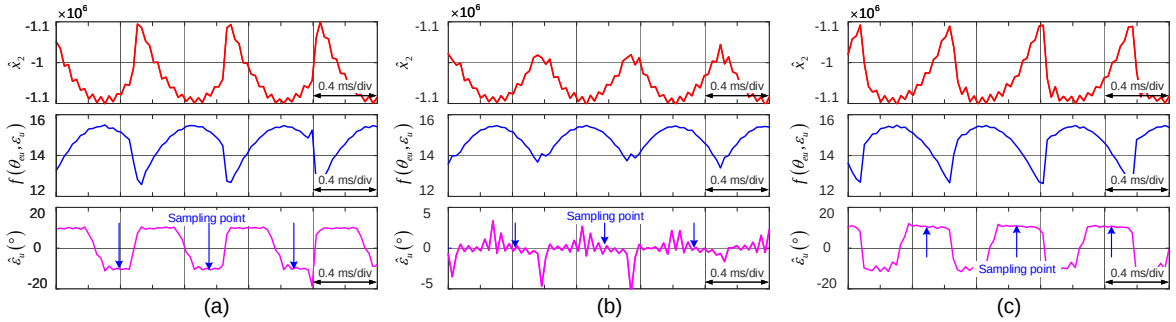


Fig. 18. The disturbance estimation of the ESO and the estimated commutation error at 9000 r/min. (a) Advanced commutation by 10°. (b) Normal commutation. (c) Lagged commutation by 10°.

the DC-link voltage and current are sampled after the Buck converter, and the Buck converter is controlled by a current controller with the DC-link current feedback. With the DC-link voltage and current, the disturbance is estimated by the ESO, and through (24), the conducting back-EMF $f(\theta_{eu}, \varepsilon_u)$ is calculated. Since $f(\theta_{eu}, \varepsilon_u)$ contains the commutation error information, according to (25), the commutation error $\hat{\varepsilon}_u$ is extracted. Then, the commutation error compensation is performed through Fig. 14.

IV. Experimental Evaluation

The proposed fast commutation error compensation method for the BLDC motor is applied on an MSCMG experiment platform, as shown in Fig. 16. The MSCMG system mainly consists of three components, an AMB system, a gimbal servo motor, and a high-speed BLDC motor. The flywheel rotor is suspended by the AMB system and then driven to rotate by the high-speed motor controller. The gimbal servo motor is used to change the direction of the gimbal to produce torque. To enhance the drive capability of the high-speed motor, a double symmetry motor in the gyro room is adopted, and the motor parameters are shown in TABLE I. To reduce the size and weight of the hardware system, the controllers of the high-speed motor and the AMB system are integrated, where a DSP+FPGA hybrid control structure is used. Since FPGA can carry on parallel operation through internal gate arrays and shows dramatic performance at doing logic and timing operations, the commutation logic is executed in FPGA. Because of DSP's powerful signal processing capabilities and abundant interface resources, the commutation error detection and compensation algorithm, the closed-loop control, and AD conversion are implemented in DSP.

A. Performance Evaluation of the ESO

The ESO is used to estimate the lumped disturbance that contains the commutation error information. A fast and high-precision ESO is critical for the commutation error estimation. Fig. 17 and Fig. 18 show the performance of the ESO with the advanced commutation, the normal commutation and the lagged commutation at 9000 r/min.

As shown in Fig. 17, x_1 is the DC-link current that has been defined as a state variable of the system, and $\hat{x}_1$ is the estimation of x_1. The ESO estimation process contains two stages, the tracking stage and the steady-state stage. It is clear that the estimated DC-link current can track the actual DC-link current quickly. As shown in the bottom sub-figures of Fig. 17, once the estimated DC-link current has tracked the actual DC-link current, the current estimation error $\hat{x}_1 - x_1$ will maintain at a small value.

To illustrate the commutation error estimation process, the estimated lumped disturbance of the ESO and the estimated commutation error are presented in Fig. 18. As presented in (19), the lumped disturbance includes the voltage drops caused by the phase resistance and the conducting back-EMF $f(\theta_{eu}, \varepsilon_u)$ that contains the commutation error information. As can be seen from Fig. 18 (b), when the commutation is ideal, $f(\theta_{eu}, \varepsilon_u)$ is symmetrical and is equal at the beginning and end of commutation. Once the commutation point deviates the ideal commutation point, advanced or lagged, $f(\theta_{eu}, \varepsilon_u)$ is not symmetrical anymore. In order to demonstrate the

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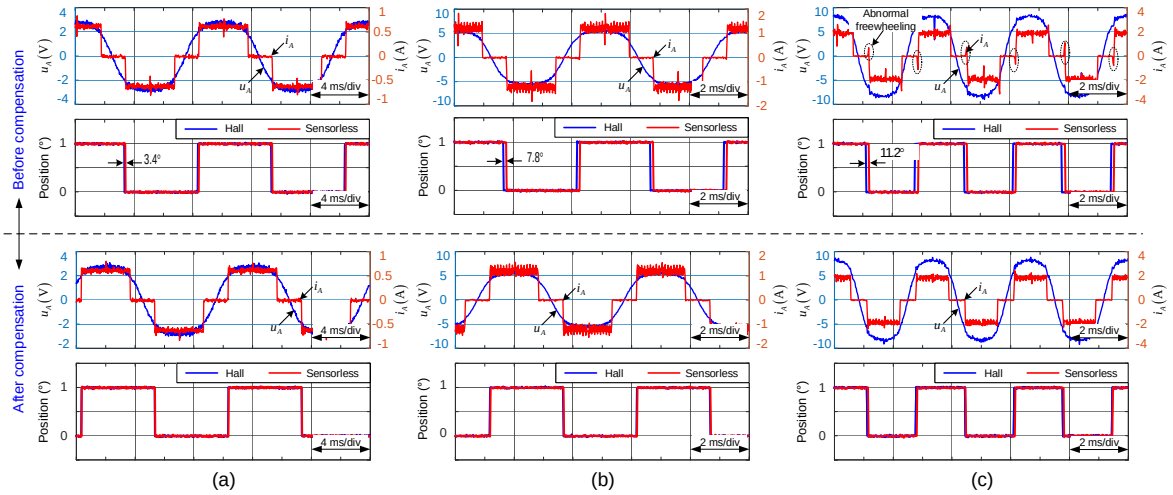


Fig. 19. The terminal voltage and the phase current with and without commutation error compensation. (a) 3000 r/min. (b) 6000 r/min. (c) 9000 r/min.

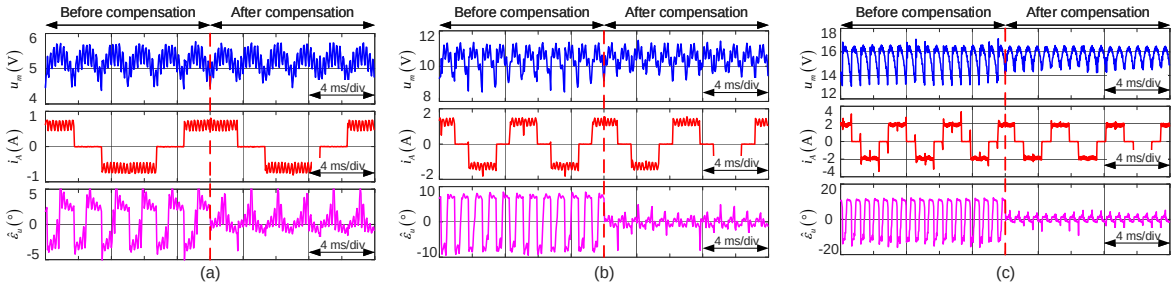


Fig. 20. Transition process of commutation error compensation. (a) 3000 r/min. (b) 6000 r/min. (c) 9000 r/min.

sampling process of commutation error, the estimated commutation error $\hat{\varepsilon}_u$ is also presented. As shown in Fig. 18, when a commutation error is added, the ESO can all estimate the commutation error. When the commutation is advanced, $\hat{\varepsilon}_u < 0$; otherwise, $\hat{\varepsilon}_u \geq 0$, which is also consistent with Fig. 13. It is noted that the ESO has very fast convergence speed, and the fluctuation of $\hat{\varepsilon}_u$ is caused by the transformation shown in Fig. 13.

B. Performance Test under Constant Speed

Since the BLDC motor of the MSCMG mainly works at a constant speed, the performance of the motor under constant speed is of vital importance. To verify the effectiveness of the proposed commutation error compensation method, experimental tests are performed at 3000 r/min, 6000 r/min and 9000 r/min, respectively. Fig. 19 (a) shows the terminal voltage and the phase current at 3000 r/min with and without commutation error compensation. As can be seen, when the motor operates at 3000 r/min, the commutation error is very small and the waveform of the phase current also looks normal. However, when the speed increases to 6000 r/min, as shown in Fig. 19 (b), the commutation error has reached to about 7.8°, and the waveform of the phase current begins to be distorted.

Benefit from the AMB system and the vacuum environment of the gyro room provided by a vacuum pump, the BLDC motor can work at rated speed with a small current. As analyzed in Fig. 7, this type of motor is more prone to the abnormal freewheeling of the diode in the non-conducting phase. When the BLDC motor operates at 9000 r/min without commutation error compensation, the commutation error has reached to 11.2°, and the phenomenon of the abnormal freewheeling in non-conducting phase is very serious, as shown in Fig. 19 (c). There is no doubt that the abnormal freewheeling will lead to negative torque and reduce motor efficiency. After performing the proposed commutation error compensation method, the abnormal freewheeling disappears and the phase current is restored to an ideal state.

C. Transient Performance Test

To verify the stability and rapidity of the proposed commutation error compensation method, the transition process is given in Fig. 20 at 3000 r/min, 6000 r/min and 9000 r/min, respectively. It can be seen from Fig. 20 that the commutation error can be estimated within only one commutation cycle, and the commutation error can also be compensated at the next commutation moment, which verifies that the proposed method has a fast compensation speed.

The transient performance of the proposed commutation error compensation method when the motor accelerates from 8900 r/min to 9000 r/min is shown in Fig. 21. As can be seen, affected by sampling error and system noise, the commutation error still has a fluctuation of about 1.8° before and after commutation error compensation. When the motor operates at high speed, various lags from the system increase, as well as the commutation error. As shown in Fig. 21, before compensation, the commutation error has reached to about 11°. After the

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proposed commutation error compensation method is adopted, the commutation error converges to zero. It is also presented in Fig. 20 that the proposed commutation error compensation method can correct the commutation error within only one commutation cycle.

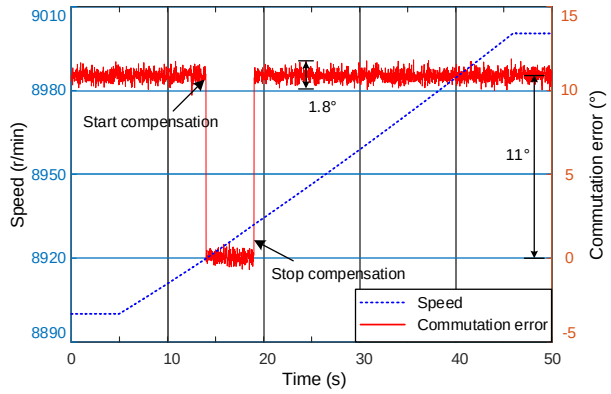


Fig. 21. Transient performance of the proposed commutation compensation method under acceleration.

D. Efficiency Comparison

In the MSCMG system, the BLDC motor control system and power drive component are powered by the same power supply whose voltage is 28 V. Fig. 22 shows the comparison of the power supply currents with and without commutation error compensation. As can be seen, with the increase in speed, power consumption increases correspondingly. The power consumption before commutation error compensation is significantly larger than that of after commutation compensation. It can be obtained from Fig. 22 that with the proposed commutation error compensation method, the power consumption can reduce 13.2% under rated speed, which verifies the significance of this study.

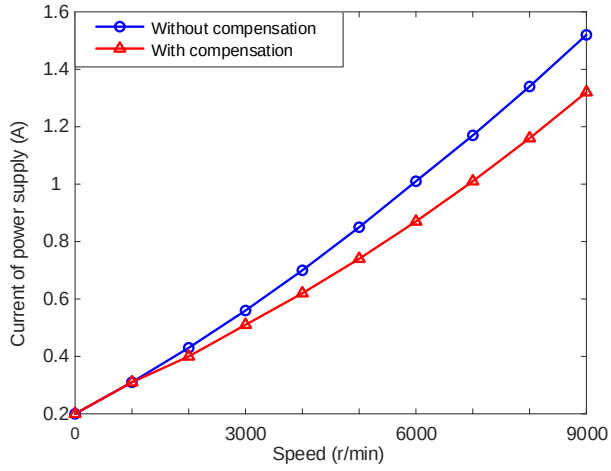


Fig. 22. Comparison of the power supply currents with and without commutation error compensation.

V. Conclusion

This paper proposed a fast commutation error compensation method for the MSCMG BLDC motor to improve the operational efficiency and reduce the losses. Combined with the characteristics of non-ideal back-EMF and the Buck converter based control topology, a novel DC-link voltage expression that contains the conducting back-EMF was derived. Since the commutation error is contained in the conducting back-EMF, the ESO was established to estimate the conducting back-EMF by considering it as a lumped disturbance. According to the estimated conducting back-EMF, it obtains the commutation error. The distinct advantages of the proposed fast commutation error compensation method are:

(1) With the continuous estimation, the commutation error can be obtained within only one commutation cycle, which guarantees the compensation speed.

(2) Because only the DC-link voltage and current are needed to perform the proposed commutation error compensation method and the double closed-loop control, it provides a preferred control strategy for motors with low-cost.

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